

حق دار

HaqDar AI

Your Rights. Your Voice. Your Proof.

Code for Pakistan: AI for Civic Innovation Hackathon 2026

Theme 2 — Technology for Civic Good

SDG 16: Peace, Justice & Strong Institutions | SDG 10: Reduced Inequalities

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Deadline: June 13, 2026 | Final Event: June 18, 2026

1. Vision & Mission

In Pakistan, the greatest weapon of institutional corruption is the silence of the ordinary citizen — forced by a system made complex and intimidating by design. HaqDar AI breaks that silence.

HaqDar AI is Pakistan's first voice-first legal rights assistant. It accepts a complaint in plain Urdu, identifies the relevant Pakistani law violated, names the responsible authority, generates a legally formatted complaint letter, and maps the complaint anonymously on a live corruption heatmap — all in under 5 seconds.

Mission: To transform every unstructured citizen voice into structured civic intelligence.

"Today, citizens struggle to prove they were wronged. HaqDar AI turns every voice note into evidence, every complaint into data, and every citizen into a participant in democracy."

2. The Problem

220 million Pakistanis have rights they cannot claim. The legal and administrative system is deliberately complex, leaving low-literacy and marginalized citizens powerless against institutional abuse.

Who suffers daily:

- Daily wage worker fined by police without receipt — no idea which law protects him
- Woman denied hospital care — doesn't know her right to emergency treatment
- Tenant harassed by landlord — no accessible legal resource in Urdu
- Citizen overcharged by WAPDA — no tool to generate a formal complaint
- Student refused admission rights — unaware of fee regulation laws

Why the problem persists:

- Legal system operates in English — inaccessible to 70% of citizens
- No free Urdu legal rights tool exists in Pakistan
- Corruption goes unreported because reporting is too complex
- Citizens don't know which authority to complain to
- Formal complaint letters require legal knowledge most people don't have

3. Solution Overview

HaqDar AI is a web-based platform (with a WhatsApp bot roadmap) that turns a simple Urdu complaint into a complete legal action package in seconds.

Core 5-Step Flow:

1. User types or speaks complaint in Urdu or English
2. AI identifies the legal violation and responsible authority
3. AI generates a formal complaint letter in legal Urdu instantly
4. Letter is routed to the correct department
5. Anonymous report added to live corruption heatmap

4. All Features — Complete Specification

Feature 1 — Core Complaint Flow (MUST HAVE)

The heart of HaqDar. User submits complaint, AI returns full legal analysis.

Input:

- Text input box with Urdu RTL support
- 4 quick-tap example complaints (pre-loaded common scenarios)
- Voice input button using Web Speech API (browser-based, no Whisper needed)

AI Output (Gemini 1.5 Flash):

- violation_summary — what happened in plain Urdu
- law_reference — exact Pakistani law name + section number
- responsible_authority — which government office to complain to
- complaint_letter — full formal letter in legal Urdu
- confidence_score — high / medium / needs verification
- citizen_rights — rights explained simply in Urdu
- evidence_to_collect — what proof to gather
- next_steps — 3 action steps in Urdu
- sdg_alignment — SDG 16 / SDG 10 / both

Pakistani Laws Pre-loaded in Prompt:

- Punjab Police Rules 2017

- Pakistan Penal Code
- Punjab Local Government Act
- Consumer Protection Act 2005
- Right of Access to Information Act 2017
- Pakistan Labour Laws
- Motor Vehicle Ordinance 1965
- Punjab Healthcare Commission Act

Feature 2 — Trust / Confidence Score

Every AI response gets a confidence rating. This shows responsible AI design to judges.

- HIGH CONFIDENCE — AI matched known law with high certainty
- MEDIUM CONFIDENCE — Partial match, citizen should verify
- NEEDS VERIFICATION — Complex case, recommend consulting a lawyer
- Color coded: Green / Yellow / Red
- One-line explanation of why this confidence level was assigned

Feature 3 — Know Your Rights Tab

Educational mode. Same Gemini call, different prompt. Turns HaqDar from reactive to proactive.

- User submits scenario (e.g. 'School mein extra fee maang rahe hain')
- AI responds: Your right in this situation
- AI responds: Exact law that protects you
- AI responds: Evidence to collect before complaining
- AI responds: 3 next steps in plain Urdu
- AI responds: Which authority is responsible
- Works as a separate tab on the same page — no new backend needed

Feature 4 — Civic Pulse Dashboard

Data visualization showing corruption trends across Pakistan. Strongly supports Open Data theme.

Dashboard sections:

- Top 5 most reported issue types this month
- District-wise complaint rankings (top 10 districts)
- Monthly trend chart — complaints over 6 months
- Issue category breakdown: Police / Utilities / Healthcare / Labour / Other
- SDG 16 progress indicator

Note: Use hardcoded realistic demo data for hackathon. Real data populates post-launch.

Feature 5 — SDG Alignment Section

Static UI section showing global impact alignment. Judges love this.

- SDG 16: Peace, Justice & Strong Institutions — badge + one line explanation
- SDG 10: Reduced Inequalities — badge + one line explanation
- Displayed on landing page and in result cards
- No backend logic required — purely UI

Feature 6 — Anonymous Heatmap

Live corruption map of Pakistan showing report density by district.

- Static map image of Pakistan with colored dots
- Dot size = number of reports in that district
- Clicking a dot shows: top issue type + count this month
- Demo data: 6-8 cities pre-populated with realistic numbers
- Bottom label: 1,247 anonymous reports this month

Feature 7 — Evidence Locker (ROADMAP)

Users upload photos, audio, documents. AI organizes into a case file.

- File upload: photos, PDFs, audio recordings
- AI reads uploaded fine/notice and says if it is legal
- Organizes evidence into timestamped case file
- Generates evidence summary for complaint

Mark as roadmap feature on Devpost.

Feature 8 — Multi-Language Support (ROADMAP)

Urdu + English + Punjabi support.

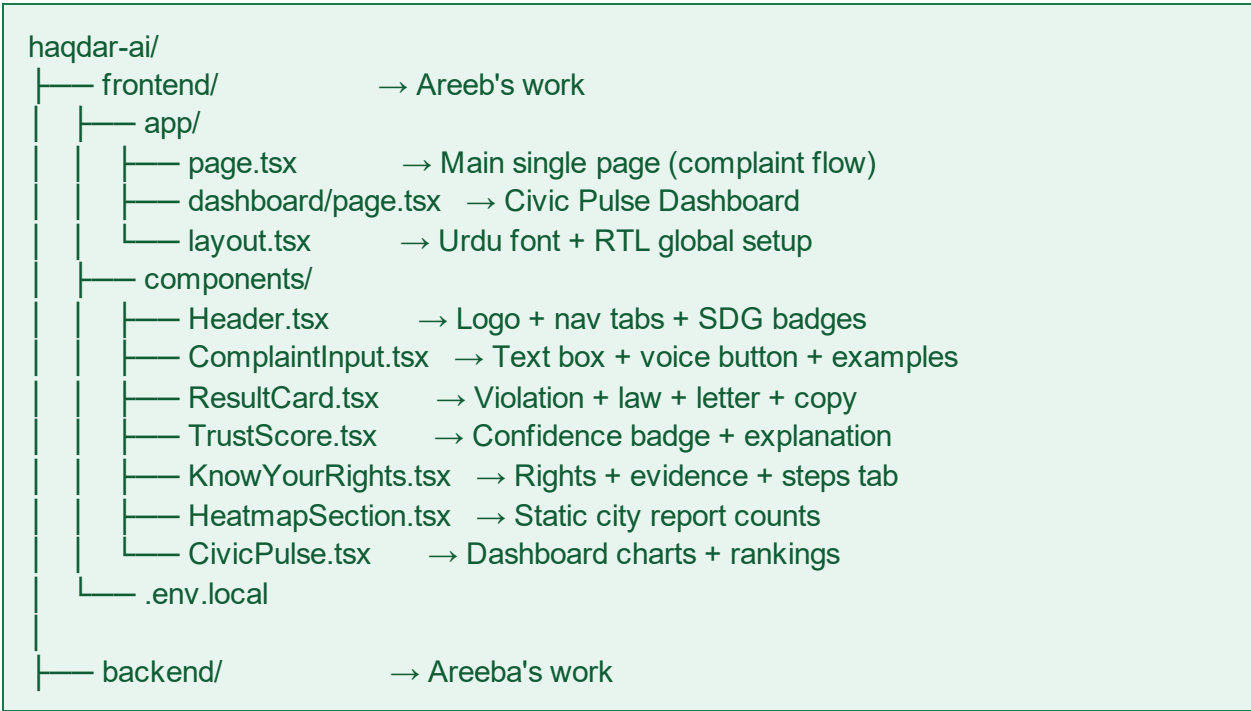
- Language switcher in header
- Translated prompts for each language
- RTL/LTR handling per language

Mark as roadmap feature on Devpost.

5. Technology Stack

Layer	Technology	Purpose	Cost
Frontend	Next.js 14 + Tailwind CSS	UI + RTL Urdu layout	Free
Backend	FastAPI (Python)	API server + routing	Free
AI Model	Google Gemini 1.5 Flash	Legal analysis + letter generation	Free tier
Voice Input	Web Speech API	Browser-based Urdu speech recognition	Free
Map	Static image + CSS dots	Corruption heatmap demo	Free
Database	None (MVP)	Stateless for hackathon	Free
Frontend Deploy	Vercel	Next.js hosting	Free
Backend Deploy	Render.com	FastAPI hosting	Free

6. Project Folder Structure



├──	main.py	→ FastAPI app + CORS
├──	gemini_service.py	→ Gemini API + prompt logic
├──	prompt_templates.py	→ Complaint prompt + rights prompt
├──	laws_reference.py	→ Pakistan laws database
├──	.env	
├──	README.md	
├──	devpost_materials/	
│	├──	pitch_deck.pdf
│	├──	screenshots/
│	└──	demo_video.mp4

7. Backend API Design

Endpoint	Method	Input	Output
POST /analyze	Complaint text	Full legal analysis JSON	Core feature
POST /rights	Scenario text	Rights + evidence + steps JSON	Know Your Rights
GET /health	None	Server status	Health check
GET /stats	None	Hardcoded dashboard data	Civic Pulse

Gemini Response JSON Shape:

{
violation_summary: string (plain Urdu explanation)
law_reference: string (e.g. Punjab Police Rules 2017, Section 12.4)
responsible_authority: string (e.g. DPO Office, WAPDA Helpline)
complaint_letter: string (full formal Urdu letter)
confidence_score: high medium needs_verification
confidence_reason: string (why this confidence level)
citizen_rights: string (simple Urdu explanation)
evidence_to_collect: string (what proof to gather)
next_steps: string[] (3 action steps in Urdu)
sdg_alignment: sdg16 sdg10 both
}

8. 48-Hour Build Timeline

Day 1 — BUILD (Today)

Time	Areeba (Backend/AI)	Areeb (Frontend)
9:00 AM	FastAPI project setup + Gemini API key configured	Next.js project created + Tailwind configured
10:00 AM	gemin_i_service.py working — test with raw complaints	Header component + RTL Urdu font loaded
11:00 AM	/analyze endpoint returning real Gemini JSON	ComplaintInput component with example buttons
12:00 PM	Test 10 different complaints — refine prompt	ResultCard component displaying all fields
1:00 PM	INTEGRATION — frontend calls backend successfully	INTEGRATION — connect to backend API
2:00 PM	/rights endpoint for Know Your Rights feature	TrustScore + KnowYourRights components
3:00 PM	/stats endpoint returning dashboard data	HeatmapSection with city report counts
4:00 PM	Full flow tested end to end — all features working	CivicPulse Dashboard page complete
5:00 PM	Deploy backend to Render.com	Deploy frontend to Vercel
6:00 PM	Devpost account created — skeleton submission live	GitHub repo clean + README drafted
8:00 PM	Bug fixes + prompt refinement	UI polish + mobile responsiveness

Day 2 — POLISH & SUBMIT

Time	Areeba	Areeb
9:00 AM	Pitch deck — slides 1-6 (problem + solution + demo)	Pitch deck — slides 7-12 (tech + impact + SDG + roadmap)
11:00 AM	Record 2-minute demo video walkthrough	Final UI polish + screenshots for Devpost
1:00 PM	Devpost final submission — all materials uploaded	GitHub final push — clean commit history
2:00 PM	Rehearse demo script 10 times	Rehearse demo script 10 times
3:00 PM	SUBMIT EVERYTHING ON DEVPOST	SUBMIT EVERYTHING ON DEVPOST

4:00 PM	Rest. You've won.	Rest. You've won.
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9. Pitch Deck — 12 Slides

Slide	Title	Content
1	Opening Statement	"In Pakistan, the greatest weapon of institutional corruption is the silence of the ordinary citizen." Full opening line. Black background. White text. Green accent on HaqDar.
2	The Problem	220M citizens. 3 bullet stories: the fined worker, the denied patient, the harassed tenant. Real faces. One stat per bullet.
3	Why It Persists	Legal system in English. No Urdu rights tool. Reporting too complex. Corruption unpunished. 4 bullets, visual icons.
4	How HaqDar Works	5-step numbered flow with icons. Clean diagram. Green arrows. Simple language.
5	Know Your Rights	Screenshot of the feature. One example: school fee complaint → rights output.
6	Trust Score + Responsible AI	Show the confidence badge. Explain why responsible AI matters. Judges love this.
7	LIVE DEMO	Run the actual app. This is your most important slide.
8	Civic Pulse Dashboard	Screenshot of the dashboard. 1,247 reports. District rankings. Monthly trend.
9	Tech Stack	Simple table: Next.js, FastAPI, Gemini 1.5 Flash, Vercel, Render. All free.
10	SDG Alignment + Impact	SDG 16 badge + SDG 10 badge. Impact numbers: 220M users, 5 seconds, zero literacy required.
11	Roadmap	Phase 1: Web app (done). Phase 2: WhatsApp bot. Phase 3: Direct ombudsman routing. Phase 4: Open dataset for journalists.
12	Closing	"HaqDar gives every Pakistani — regardless of education, income, or status — the power to claim what is already theirs by law. حق دار."

10. Demo Script — Memorize Word for Word

"Let me show you HaqDar in action."

"A citizen types: 'Police ne mujhe bina wajah fine diya aur receipt nahi di'"

"In 4 seconds — HaqDar identifies this violates Punjab Police Rules 2017, Section 12.4."

"The responsible authority: DPO Office."

"And here — a complete, formal complaint letter in legal Urdu, ready to submit."

"Confidence Score: HIGH — AI matched this to a known law with certainty."

"One click to copy. This citizen went from helpless to legally armed in 4 seconds."

"Now click Know Your Rights — same complaint, educational mode."

"Your right. Your evidence. Your next steps. All in plain Urdu."

"Now look at the Civic Pulse Dashboard."

"34 similar reports in this district this month alone."

"Police complaints up 23% in Rawalpindi this month."

"HaqDar doesn't just help one citizen. It exposes the pattern."

"That is how you hold a system accountable."

11. Judge Q&A — Prepared Answers

Q: Is the legal information accurate?

"We use Gemini with a curated prompt grounded in real Pakistani law references including Punjab Police Rules 2017, Consumer Protection Act, and the Right to Information Act 2017. We recommend users verify with a lawyer for high-stakes cases — our goal is awareness and access, not legal advice replacement."

Q: What about data privacy?

"All reports are anonymous by default. No personal information is stored. The heatmap shows only aggregated patterns — no individual report is identifiable."

Q: How do you scale this?

"Phase 2 is a WhatsApp bot — zero app download needed. Pakistan has 50M+ WhatsApp users. Roman Urdu NLU makes it work for every demographic. That is our distribution channel."

Q: Why not just Google the law?

"A daily wage worker in Gujranwala does not know which law was violated, which office to contact, or how to write a formal letter in legal Urdu. Google gives 10 blue links. HaqDar gives one complete answer in 4 seconds in his own language."

Q: What makes this different from existing legal aid apps?

"No tool in Pakistan combines Urdu voice input, automatic law detection, formal complaint letter generation, and anonymous corruption mapping in a single flow. We are not a directory. We are an action engine."

12. Devpost Submission Content

Project Title:

HaqDar AI — حق دار

Tagline:

Pakistan's first AI legal rights assistant. Speak your complaint. Know your rights. Get your letter.

Inspiration:

Millions of Pakistanis face illegal fines, service denials, and institutional harassment every day. The system's complexity silences them by design. HaqDar was built to break that silence — to give every citizen, regardless of education or income, the power to claim what is already theirs by law.

What it does:

HaqDar accepts a complaint in plain Urdu, identifies the relevant Pakistani law violated, names the responsible authority, generates a legally formatted complaint letter, provides a Know Your

Rights education panel, assigns a Trust/Confidence Score, and maps the complaint anonymously on a corruption heatmap.

How we built it:

Next.js 14 frontend with Tailwind CSS and RTL Urdu support. FastAPI backend. Google Gemini 1.5 Flash for legal analysis and complaint generation. Deployed on Vercel (frontend) and Render (backend). All free tier.

Challenges:

Prompt engineering for consistent JSON output in legal Urdu. RTL layout with Tailwind. Making the Gemini response accurate for Pakistani-specific laws.

Accomplishments:

Working deployed app in 48 hours. Urdu letter generation that reads as genuinely formal and legally appropriate. Demo that creates an emotional response in under 90 seconds.

What's next:

WhatsApp bot with Roman Urdu NLU. Direct routing to FIA and provincial ombudsman portals. Open anonymized corruption dataset for journalists and researchers. Evidence Locker for document uploads.

SDG Alignment:

SDG 16 — Peace, Justice & Strong Institutions. SDG 10 — Reduced Inequalities.

Theme:

Theme 2 — Technology for Civic Good

13. Why HaqDar Wins

Other Teams	HaqDar AI
Figma mockups called working prototypes	Deployed working app with real API
15 features, none complete	5 features, all polished
English-only interface for Pakistani citizens	Urdu RTL with voice input
Generic civic dashboards	Specific legal violation detection
Forgettable project names	حق دار — judges remember it
No responsible AI consideration	Trust Score shows ethical design
No global alignment	SDG 16 + SDG 10 explicitly addressed
Features list as roadmap	Working demo that makes judges feel something

"HaqDar gives every Pakistani — regardless of education, income, or status — the power to claim what is already theirs by law."

حق دار

Built to win. Built for Pakistan.